

Лабораторная работа №7

Операционные системы

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Информация

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Цель работы

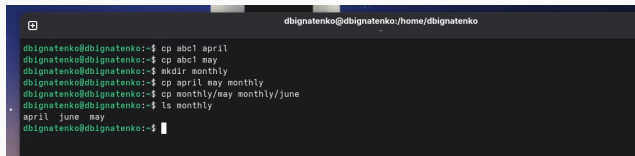
Ознакомление с файловой системой Linux, её структурой. Приобретение практических навыков по применению команд для работы с файлами и каталогами.

Задание

- Выполнить примеры из описания лабораторной работы
- Выполнить задания по копированию и перемещению файлов
- Определить опции `chmod` для установки прав доступа
- Изучить `man`-страницы команд `mount`, `fsck`, `mkfs`, `kill`

Выполнение лабораторной работы

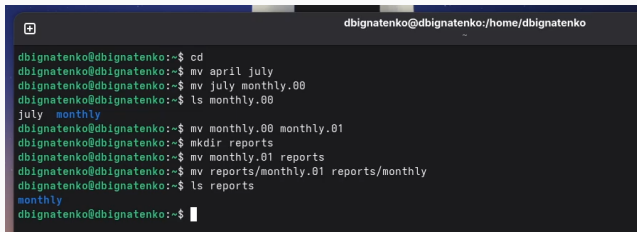
Копирую файлы с помощью команды `cp`, создаю каталог `monthly`.

A terminal window with a dark background and light text. The title bar at the top reads 'dbignatenko@dbignatenko: /home/dbignatenko'. The terminal shows a series of commands and their outputs. The commands are: 'cp abc1 april', 'cp abc1 may', 'mkdir monthly', 'cp april may monthly', 'cp monthly/may monthly/june', and 'ls monthly'. The outputs are: 'april june may' and a cursor on a new line.

```
dbignatenko@dbignatenko:~$ cp abc1 april
dbignatenko@dbignatenko:~$ cp abc1 may
dbignatenko@dbignatenko:~$ mkdir monthly
dbignatenko@dbignatenko:~$ cp april may monthly
dbignatenko@dbignatenko:~$ cp monthly/may monthly/june
dbignatenko@dbignatenko:~$ ls monthly
april june may
dbignatenko@dbignatenko:~$
```

Рис. 1: Копирование файлов

Переименовываю файлы и каталоги с помощью команды mv.

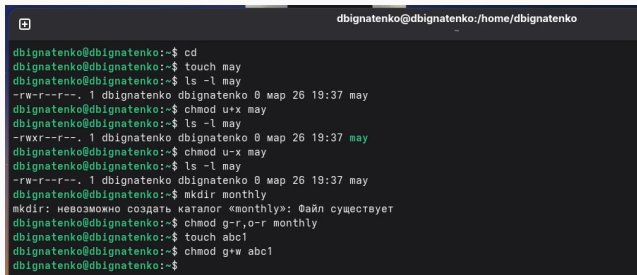
A terminal window with a dark background and light-colored text. The window title bar shows a plus icon and the text 'dbignatenko@dbignatenko:/home/dbignatenko'. The terminal content shows a series of commands and their outputs. The commands are: 'cd', 'mv april july', 'mv july monthly.00', 'ls monthly.00', 'mv monthly.00 monthly.01', 'mkdir reports', 'mv monthly.01 reports', 'mv reports/monthly.01 reports/monthly', 'ls reports', and 'monthly'. The outputs are: 'july monthly', 'monthly.00', 'reports', and 'reports/monthly'. The prompt 'dbignatenko@dbignatenko:~\$' is visible at the end of each line.

```
dbignatenko@dbignatenko:~$ cd
dbignatenko@dbignatenko:~$ mv april july
dbignatenko@dbignatenko:~$ mv july monthly.00
dbignatenko@dbignatenko:~$ ls monthly.00
july  monthly
dbignatenko@dbignatenko:~$ mv monthly.00 monthly.01
dbignatenko@dbignatenko:~$ mkdir reports
dbignatenko@dbignatenko:~$ mv monthly.01 reports
dbignatenko@dbignatenko:~$ mv reports/monthly.01 reports/monthly
dbignatenko@dbignatenko:~$ ls reports
monthly
dbignatenko@dbignatenko:~$
```

Рис. 2: Перемещение и переименование

Изменение прав доступа

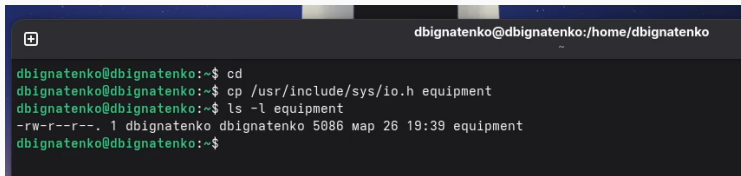
Изменяю права доступа с помощью команды `chmod`.



```
dbignatenko@dbignatenko:/home/dbignatenko
dbignatenko@dbignatenko:~$ cd
dbignatenko@dbignatenko:~$ touch may
dbignatenko@dbignatenko:~$ ls -l may
-rw-r--r--. 1 dbignatenko dbignatenko 0 map 26 19:37 may
dbignatenko@dbignatenko:~$ chmod u+x may
dbignatenko@dbignatenko:~$ ls -l may
-rwxr--r--. 1 dbignatenko dbignatenko 0 map 26 19:37 may
dbignatenko@dbignatenko:~$ chmod u-x may
dbignatenko@dbignatenko:~$ ls -l may
-rw-r--r--. 1 dbignatenko dbignatenko 0 map 26 19:37 may
dbignatenko@dbignatenko:~$ mkdir monthly
mkdir: невозможно создать каталог «monthly»: Файл существует
dbignatenko@dbignatenko:~$ chmod g-r,o-r monthly
dbignatenko@dbignatenko:~$ touch abc1
dbignatenko@dbignatenko:~$ chmod g+w abc1
dbignatenko@dbignatenko:~$
```

Рис. 3: Примеры `chmod`

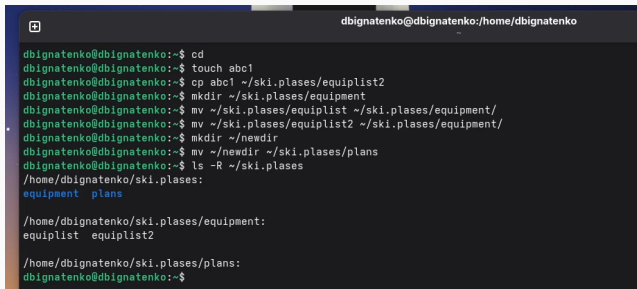
Копирую файл `/usr/include/sys/io.h` в `equipment`.

A terminal window with a dark background and light green text. The title bar at the top reads 'dbignatenko@dbignatenko:/home/dbignatenko'. The terminal shows a sequence of commands and their output: 'cd' changes the directory to '~'; 'cp /usr/include/sys/io.h equipment' copies the file; 'ls -l equipment' lists the file with permissions '-rw-r--r--', owner 'dbignatenko', group 'dbignatenko', size '5086', and timestamp 'map 26 19:39'.

```
dbignatenko@dbignatenko:~$ cd
dbignatenko@dbignatenko:~$ cp /usr/include/sys/io.h equipment
dbignatenko@dbignatenko:~$ ls -l equipment
-rw-r--r--. 1 dbignatenko dbignatenko 5086 map 26 19:39 equipment
dbignatenko@dbignatenko:~$
```

Рис. 4: Копирование `io.h`

Создаю структуру каталогов и проверяю её командой `ls -R`.



```
dbignatenko@dbignatenko:~/home/dbignatenko
dbignatenko@dbignatenko:~$ cd
dbignatenko@dbignatenko:~$ touch abc1
dbignatenko@dbignatenko:~$ cp abc1 ~/ski.plases/equiplist2
dbignatenko@dbignatenko:~$ mkdir ~/ski.plases/equipment
dbignatenko@dbignatenko:~$ mv ~/ski.plases/equiplist ~/ski.plases/equipment/
dbignatenko@dbignatenko:~$ mv ~/ski.plases/equiplist2 ~/ski.plases/equipment/
dbignatenko@dbignatenko:~$ mkdir ~/newdir
dbignatenko@dbignatenko:~$ mv ~/newdir ~/ski.plases/plans
dbignatenko@dbignatenko:~$ ls -R ~/ski.plases
/home/dbignatenko/ski.plases:
equipment  plans

/home/dbignatenko/ski.plases/equipment:
equiplist  equiplist2

/home/dbignatenko/ski.plases/plans:
dbignatenko@dbignatenko:~$
```

Рис. 5: Структура ski.plases

chmod 744 australia

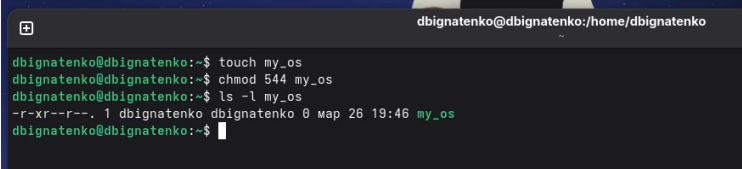
Устанавливаю права drwxr-r- для каталога australia.

```
dbignatenko@dbignatenko:~/home/dbignatenko
dbignatenko@dbignatenko:~$ mkdir australia
dbignatenko@dbignatenko:~$ chmod 744 australia
dbignatenko@dbignatenko:~$ ls -l
итого 36
-rw-rw-r--. 1 dbignatenko dbignatenko  0 мар 26 19:42 abc1
drwxr--r--. 1 dbignatenko dbignatenko  0 мар 26 19:46 australia
drwxr-xr-x. 1 dbignatenko dbignatenko  0 мар  9 15:53 bin
drwxr-xr-x. 1 dbignatenko dbignatenko 468 мар  6 19:47 den4ik-dev.github.io
drwxr-xr-x. 1 dbignatenko dbignatenko  0 мар 11 01:17 Documents
drwxr-xr-x. 1 dbignatenko dbignatenko  0 мар 11 01:17 Downloads
drwxr-xr-x. 1 dbignatenko dbignatenko  76 мар  6 00:31 git-extended
-rw-r--r--. 1 dbignatenko dbignatenko 18657 мар  9 15:55 LICENSE
-rw-r--r--. 1 dbignatenko dbignatenko  0 мар 26 19:37 may
drwx-x--x. 1 dbignatenko dbignatenko  24 мар 26 19:33 monthly
-rw-r--r--. 1 dbignatenko dbignatenko 12892 фев 25 19:37 README.md
drwxr-xr-x. 1 dbignatenko dbignatenko  14 мар 26 19:36 reports
drwxr-xr-x. 1 dbignatenko dbignatenko  28 мар 26 19:43 ski.places
drwxr-xr-x. 1 dbignatenko dbignatenko  10 мар  2 00:38 work
drwxr-xr-x. 1 dbignatenko dbignatenko 142 мар 13 00:22 Видео
drwxr-xr-x. 1 dbignatenko dbignatenko  0 мар  1 21:51 Документы
drwxr-xr-x. 1 dbignatenko dbignatenko 402 мар  9 15:37 Загрузки
drwxr-xr-x. 1 dbignatenko dbignatenko  50 мар  4 01:36 Изображения
drwxr-xr-x. 1 dbignatenko dbignatenko  0 мар  1 21:51 Музыка
drwxr-xr-x. 1 dbignatenko dbignatenko  0 мар  1 21:51 Общедоступные
drwxr-xr-x. 1 dbignatenko dbignatenko  0 мар  1 21:51 'Рабочий стол'
drwxr-xr-x. 1 dbignatenko dbignatenko  0 мар  1 21:51 Шаблоны
dbignatenko@dbignatenko:~$
```

Рис. 6: chmod 744 australia

chmod 544 my_os

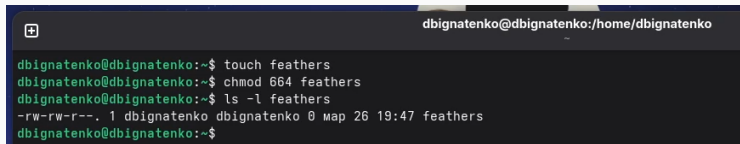
Устанавливаю права -r-xr-r- для файла my_os.

A terminal window with a dark background and light green text. The title bar at the top reads "dbignatenko@dbignatenko:/home/dbignatenko". The terminal shows a sequence of commands and their output: "touch my_os", "chmod 544 my_os", and "ls -l my_os". The output of the last command is "-r-xr-r--. 1 dbignatenko dbignatenko 0 map 26 19:46 my_os".

```
dbignatenko@dbignatenko:~$ touch my_os
dbignatenko@dbignatenko:~$ chmod 544 my_os
dbignatenko@dbignatenko:~$ ls -l my_os
-r-xr-r--. 1 dbignatenko dbignatenko 0 map 26 19:46 my_os
dbignatenko@dbignatenko:~$
```

Рис. 7: chmod 544 my_os

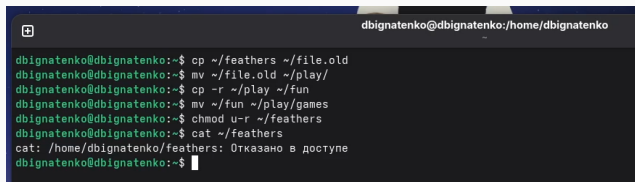
Устанавливаю права -rw-rw-r- для файла feathers.

A terminal window with a dark background and light green text. The title bar at the top shows a plus icon and the path 'dbignatenko@dbignatenko:/home/dbignatenko'. The terminal contains four lines of text: a prompt followed by 'touch feathers', a prompt followed by 'chmod 664 feathers', a prompt followed by 'ls -l feathers', and the output of the last command: '-rw-rw-r--. 1 dbignatenko dbignatenko 0 map 26 19:47 feathers'.

```
dbignatenko@dbignatenko:~$ touch feathers
dbignatenko@dbignatenko:~$ chmod 664 feathers
dbignatenko@dbignatenko:~$ ls -l feathers
-rw-rw-r--. 1 dbignatenko dbignatenko 0 map 26 19:47 feathers
dbignatenko@dbignatenko:~$
```

Рис. 8: chmod 664 feathers

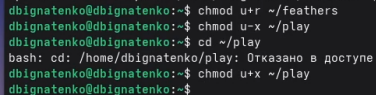
При отсутствии права на чтение получаю ошибку “Отказано в доступе”.



```
dbignatenko@dbignatenko:/home/dbignatenko
dbignatenko@dbignatenko:~$ cp ~/feathers ~/file.old
dbignatenko@dbignatenko:~$ mv ~/file.old ~/play/
dbignatenko@dbignatenko:~$ cp -r ~/play ~/fun
dbignatenko@dbignatenko:~$ mv ~/fun ~/play/games
dbignatenko@dbignatenko:~$ chmod u-r ~/feathers
dbignatenko@dbignatenko:~$ cat ~/feathers
cat: /home/dbignatenko/feathers: Отказано в доступе
dbignatenko@dbignatenko:~$
```

Рис. 9: Ошибки Permission denied

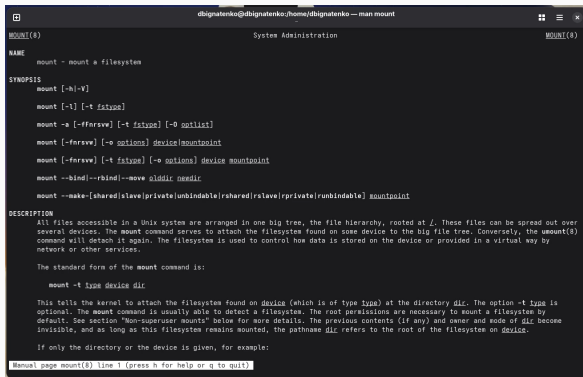
При отсутствии права на выполнение нельзя войти в каталог.



```
dbignatenko@dbignatenko:~$ chmod u+r ~/feathers
dbignatenko@dbignatenko:~$ chmod u-x ~/play
dbignatenko@dbignatenko:~$ cd ~/play
bash: cd: /home/dbignatenko/play: Отказано в доступе
dbignatenko@dbignatenko:~$ chmod u+x ~/play
dbignatenko@dbignatenko:~$
```

Рис. 10: Ошибка cd в каталог

Команда mount монтирует файловые системы.



```
dbignatenko@dbignatenko:/home/dbignatenko -- man mount
MOUNT(8)                                System Administration                                MOUNT(8)

NAME
    mount - mount a filesystem

SYNOPSIS
    mount [-h|-V]

    mount [-t] [-t fstype]

    mount -a [-ffnrsvw] [-t fstype] [-O optlist]

    mount [-fnrsvw] [-o options] device/mountpoint

    mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

    mount --bind|--rbind|--move olddir newdir

    mount --make=[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION
    All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at /. These files can be spread out over several devices. The mount command serves to attach the filesystem found on some device to the big file tree. Conversely, the umount(8) command will detach it again. The filesystem is used to control how data is stored on the device or provided in a virtual way by network or other services.

    The standard form of the mount command is:

        mount -t type device dir.

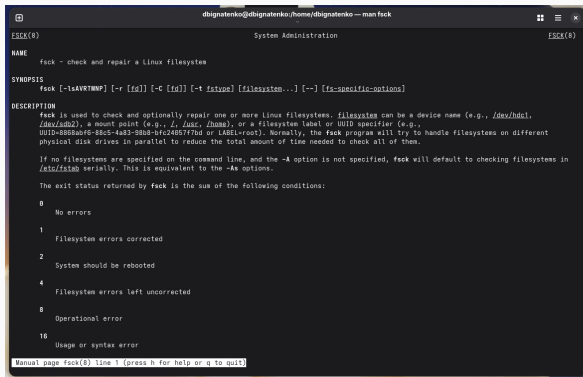
    This tells the kernel to attach the filesystem found on device (which is of type type) at the directory dir. The option -t type is optional. The mount command is usually able to detect a filesystem. The root permissions are necessary to mount a filesystem by default. See section "Non-superuser mounts" below for more details. The previous contents (if any) and mode of dir become invisible, and as long as this filesystem remains mounted, the pathname dir refers to the root of the filesystem on device.

    If only the directory or the device is given, for example:

    Manual page mount(8) line 1 (press h for help or q to quit)
```

Рис. 11: man mount

Команда fsck проверяет целостность файловой системы.



```
dbignatenko@dbignatenko:/home/dbignatenko — man fsck
FSCK(8)                                     System Administration          FSCK(8)

NAME
    fsck - check and repair a linux filesystem

SYNOPSIS
    fsck [-lsAVRTWNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more linux filesystems. filesystem can be a device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24857f7bd or LABEL=root). Normally, the fsck program will try to handle filesystems on different physical disk drives in parallel to reduce the total amount of time needed to check all of them.

    If no filesystems are specified on the command line, and the -A option is not specified, fsck will default to checking filesystems in /etc/fstab serially. This is equivalent to the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0      No errors

    1      Filesystem errors corrected

    2      System should be rebooted

    4      Filesystem errors left uncorrected

    8      Operational error

    16     Usage or syntax error

Manual page fsck(8) line 1 (press h for help or q to quit)
```

Рис. 12: man fsck

Выводы

В ходе выполнения лабораторной работы ознакомился с файловой системой Linux. Приобрёл практические навыки по применению команд `cp`, `mv`, `mkdir`, `chmod`. Изучил систему прав доступа и команды обслуживания файловой системы.